

Multiple Linear Regression Project

The purpose of this project is to fit a multiple regression model for a dataset.

This project has two deadlines.

Step 1

Find or collect your data. By **Sunday, April 12**, please turn in the following:

- (a) The name and source of your dataset.
- (b) A brief description of the variables you intend to use as your explanatory and response variables

Your data should consist of information on one quantitative response variable and at least two explanatory variables that you think may help predict the response variable. The explanatory variables should either be quantitative or categorical with only two categories. The sample size should be at least $n = 25$. You can use published data or collect your own.

Step 2

Carry out a complete multiple regression analysis, using the grading rubric below as a guide to what should be included. Discuss what each graph, table, and statistics inference procedure tells you about your dataset. Your report should not be just a list of equations, graphs, and computer output-rather, it should be a coherent narrative with complete sentences and paragraphs. **Include all relevant R output as part of your report. Include each graph and table at the point where you discuss it; that is do not collect all the output at the end of the report.** The final version of the project report is due on **Sunday, April 26**. A detailed grading rubric for the final draft is given below.

Grading rubric for the final draft:

- The report is neatly typed in Word. (2 points)
- The report is carefully proofread, so there are no spelling or punctuation errors. (2 points)
- The report is clearly written and interesting, with no run-on sentences, sentence fragments, or other grammatical errors. (2 points)
- The dataset, variables and questions of interest are described clearly. (3 points)
- A scatterplot is drawn for each explanatory variable. The appearance of each scatterplot is described (3 points)
- Correlations between each pair of variables are calculated and interpreted clearly in the context of the dataset. (3 points)
- An appropriate population multiple regression model is written to describe the relationship among the variables, and the assumptions for statistical inference are clearly stated. (2 points)
- The parameters of the population multiple regression model are estimated using R, and the fitted regression equation is reported. (2 points)

- Appropriate residual plots are drawn using R. The appearance of each residual plot is described. What does each plot tell you about the appropriateness of the regression model? (3 points)
- The analysis of variance F test is carried out to assess the usefulness of the multiple regression model. All components of a hypothesis test are included (hypotheses, test statistic, numerator and denominator degrees of freedom, P-value, and a conclusion in the context of the dataset) (3 point)
- R^2 for the multiple regression is reported and interpreted clearly in the context of the dataset. (2 points)
- Appropriate t tests are carried out for each of β . All components of each hypothesis test are included (hypotheses, test statistic, degrees of freedom, P-value and a conclusion in the context of the dataset). (4 points)
- The estimates of all β s in the multiple regression model (intercept and slopes) are interpreted clearly in the context of the dataset. If it doesn't make sense to interpret a particular estimate, explain why. (6 points)
- The report ends with a summary of the most important findings. (3 points)